

Prapti Maitra

praptimaitra@gmail.com · linkedin.com/in/prapti-maitra · Delhi NCR, India · Open to remote & hybrid roles

Edinburgh CS & Mathematics graduate with hands-on experience across data science, full-stack development, and ML research. Extended an open-source in-database ML framework alongside PhD researchers; led statistical analysis across 888 data points producing novel empirical findings. Built production systems independently — CRM, Power BI clinical dashboards, deep learning app at 97.7% accuracy, and a cross-platform mobile app with real-time database and maps integration. Seeking analytical, data, or product-adjacent roles where mathematical rigour and technical depth drive real decisions.

EDUCATION

University of Edinburgh

Sep 2019 – Jun 2024 | Edinburgh, UK

BSc (Hons) Computer Science & Mathematics · 2:1 Honours

EXPERIENCE

ITSec Middle East — IT Analyst

Jan – Apr 2026 | Doha, Qatar

IT Analyst

- Installed the company's first physical firewall device and produced the full network diagram; deployed endpoint security across all office devices for the company's first client.
- Developed an Information Security Management System (ISMS) using the ISMS Online platform, aligned to Qatar government cybersecurity standards — packaged as a client-facing service offering.
- Continued to maintain and expand the CRM system while managing business development contacts and supporting ongoing client relationship management.

ITSec Middle East — Cybersecurity Intern

Aug – Dec 2025 | Doha, Qatar

Cybersecurity Intern

- Built a production CRM system from scratch in TypeScript, Node.js, and Prisma ORM; sat in client intake meetings and translated business requirements into technical specifications.
- Automated internal workflows by writing Python scripts invoking the OpenAI Whisper API — reducing manual transcription and communication overhead significantly.
- Completed Sophos Certified Firewall Engineer certification and managed day-to-day company communications throughout the internship.

Synergy Gulf Trading WLL

Mar – Aug 2025 | Doha, Qatar

Freelance Web Developer

- Rebuilt company website from scratch using JavaScript — responsive design, UX optimisation, and performance improvements across all device sizes.
- Managed full client engagement independently: requirements gathering, iterative design feedback, and final delivery against agreed specifications.
- Improved site usability and visual consistency, directly aligning the platform with the client's brand identity and business goals.

Primark

Nov – Dec 2024 | Edinburgh, UK

Retail Assistant — Holiday Hire

- Hired as seasonal holiday staff during peak Christmas period; selected as the only holiday hire assigned cashier duty after demonstrating quick learning on the till.
- Managed high-volume customer transactions and handled customer queries on the shop floor, maintaining service standards during one of the busiest retail periods of the year.
- Operated effectively in a fast-paced team environment, adapting quickly to store procedures and contributing to smooth peak-season operations from day one.

Healync Technologies

May – Aug 2022 | Kolkata, India

Data Science Intern

- Built interactive Power BI dashboards visualising patient health metrics for clinical end users — adopted by client doctors to track KPIs in real time.
- Extracted and analysed 500+ records from MongoDB; identified data quality issues and automated reporting workflows, reducing project delivery time by 2 weeks.
- Facilitated cross-functional communication between engineering and clinical teams, translating technical outputs into actionable clinical insights.

University of Edinburgh — Informatics Tutoring Scheme

Dec 2021 – Apr 2022 | Edinburgh, UK

CS Tutor

- Delivered one-to-one Computer Science tutoring to National 5 students via Microsoft Teams, covering core concepts and exam preparation strategies.
- Adapted teaching approach to individual learning styles; consistently received positive feedback on clarity and student confidence improvement.
- Developed structured revision materials covering algorithms, data representation, and computational thinking to support independent study.

RESEARCH & PROJECTS

Street Dog Foster Care App · React Native · Expo · TypeScript · Supabase · PostgreSQL

2026

Independent Project · Delhi NCR NGO (in development)

- Built a cross-platform mobile app in React Native, Expo, and TypeScript to track 50+ street dogs across Delhi colonies — managing health records, vaccinations, sterilisation status, and GPS locations.
- Designed a PostgreSQL schema (Supabase) with 6 relational tables, row-level security policies, and auto-generated medical reminder schedules based on dog age and medical history.
- Integrated GPS capture, camera/photo library, push notifications with badge counts, SheetJS Excel export, and a Google Maps screen with colour-coded health-status pins per dog.

CRM System · TypeScript · Node.js · Prisma ORM

2025

- Independently architected and built a full production CRM system from scratch for a cybersecurity consultancy's client and case management workflows.
- Designed relational data models in Prisma ORM with PostgreSQL; implemented RESTful API endpoints in Node.js for full CRUD operations across all entities.
- Delivered end-to-end without supervision — from requirements analysis and schema design through to deployment and handover.

OpenDBML — In-Database Machine Learning Framework · Python

2023 – 2024

Undergraduate Researcher · School of Informatics, UoE · Supervised by Dr Amir Shaikhha

- Extended open-source Python framework to unify LibFM, LMFAO, and Morpheus — three state-of-the-art in-database ML systems — into a single accessible API.
- Delivered full Morpheus data transformation support; fixed LMFAO compatibility across 3 previously failing datasets; integrated PCA, correlation analysis, and feature engineering tools.
- Benchmarked all three systems across 8 real-world datasets; produced quantitative runtime vs accuracy analysis informing system selection recommendations.

Post-Covid UK Mathematics Assessment Study · Python · Jupyter

2023 – 2024

Mathematics Research Project · School of Mathematics, UoE · Grade: 84%

- Co-authored empirical study replicating Iannone & Simpson's longitudinal research; led all data collection and analysis across 888 modules at 24 UK universities.
- Applied 8 statistical methods including ANOVA, Mann-Whitney U, PCA, and K-means; confirmed significant post-pandemic shift in assessment practice ($p < 0.00001$).
- Produced novel empirical findings on the lasting impact of Covid-19 on UK higher education assessment — addressing a documented gap in academic literature.

Human Activity Recognition App · TensorFlow · bi-LSTM · CNN

2023

- Designed and trained a deep learning model using a hybrid CNN + Bidirectional LSTM architecture to classify human activity from sensor data.
- Achieved 97.7% accuracy on live sensor data — demonstrating strong command of sequence modelling and temporal feature extraction.
- Implemented full ML pipeline: data preprocessing, model architecture design, training, evaluation, and inference on real-time input streams.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · Swift · SQL · R
ML & Data: TensorFlow · scikit-learn · Pandas · NumPy · Power BI · Jupyter · CNN · bi-LSTM · PCA · K-means
Databases: PostgreSQL · MongoDB · Prisma ORM · Supabase · Node.js · REST APIs · Git
Statistical Methods: Regression · ANOVA · Hypothesis Testing · Probability · Correlation · Experimental Design

CERTIFICATIONS

SAS Programming · Introduction to Machine Learning · Introduction to DevOps · Sophos Certified Firewall Engineer · JP Morgan Software Engineering Job Simulation

LEADERSHIP & ACTIVITIES

Edinburgh Award for Leadership · Maths Outreach Team Leader · University of Edinburgh — Sep 2021 – Apr 2023
supervised 20 students, coordinated activities for children aged 6–17; 40% increase in participation across 5+ events.

Edinburgh Hoppers Committee · Represented gender minorities in Informatics; organised tech workshops, virtual hackathons, and industry talks. 2021–2024

Hack the Burgh 2022 · Organising committee, Scotland's largest annual student hackathon. 2022

TED Talk Edinburgh · Speaker Liaison — managed speaker coordination and logistics. 2022–2023

Classical Odissi Dancer · 7 years of training under the same guru since age 6.